Project-2: Statistical Descriptive Statistics

Author: Ali Can

This mini project aims to conduct a statistical descriptive analysis based on dataset "ToothGrowth"

0. Import

```
suppressPackageStartupMessages({
  library(tidyverse)
  library(e1071)
  library(car)
})
```

1. Data Exploration and Manipulation

1. Call the dataset and see the first 6 rows

```
data(ToothGrowth)
tg <- ToothGrowth
head(tg)
```

```
##    len supp dose
## 1  4.2   VC  0.5
## 2 11.5   VC  0.5
## 3  7.3   VC  0.5
## 4  5.8   VC  0.5
## 5  6.4   VC  0.5
## 6 10.0   VC  0.5
```

2. See the dimensions and structure of the dataset

```
cat(dim(tg))
```

```
## 60 3
```

```
str(tg)
```

```
## 'data.frame':    60 obs. of  3 variables:
##  $ len : num  4.2 11.5 7.3 5.8 6.4 10 11.2 11.2 5.2 7 ...
##  $ supp: Factor w/ 2 levels "OJ","VC": 2 2 2 2 2 2 2 2 2 2 ...
##  $ dose: num  0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 ...
```

3. See the numerical summary of the dataset

```
summary(tg)
```

```
##       len        supp         dose
##  Min.   : 4.20   OJ:30   Min.   :0.500
##  1st Qu.:13.07   VC:30   1st Qu.:0.500
##  Median :19.25           Median :1.000
```

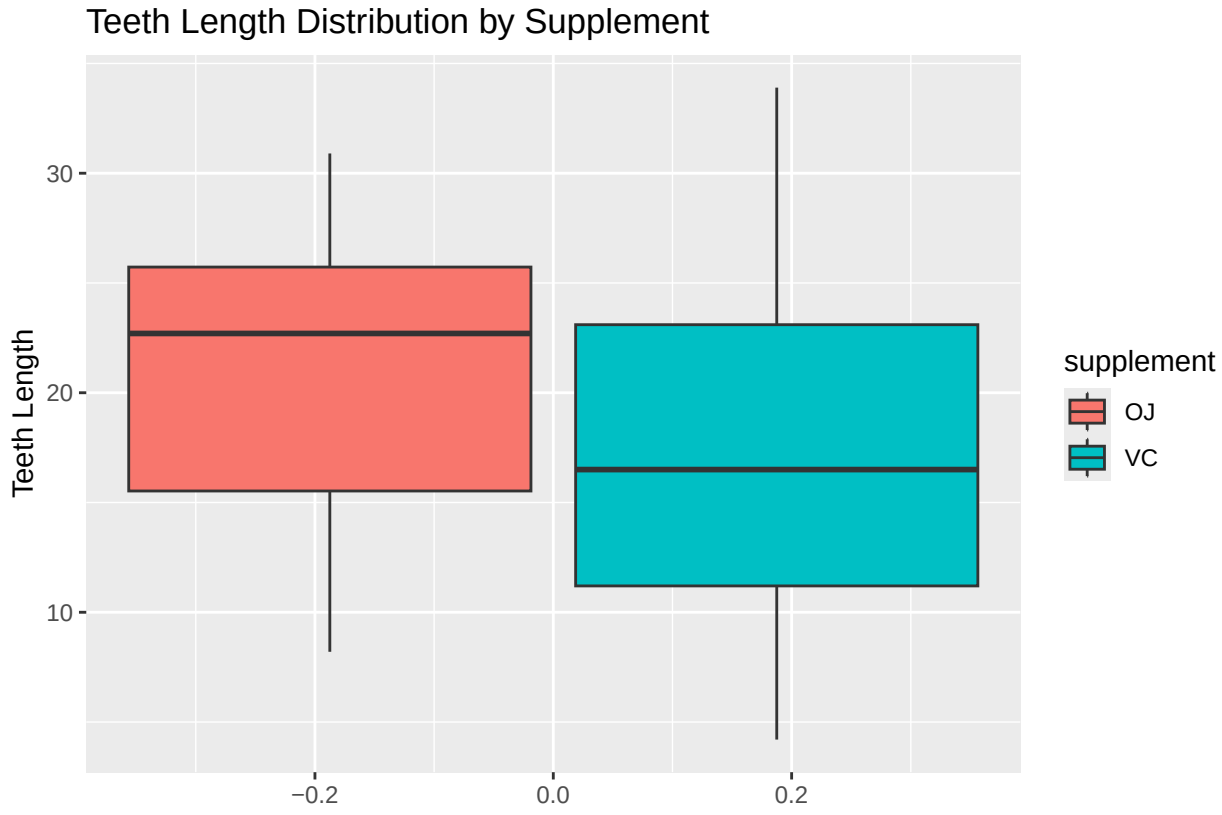
```
##  Mean   :18.81           Mean   :1.167
##  3rd Qu.:25.27           3rd Qu.:2.000
##  Max.   :33.90           Max.   :2.000
```

Obviously, dataset has 3 variables.

- "len" shows the elongation in teeth. Mean and median are close indicating normality.
- "supp" shows the method of vitamin C intake: either by pill (VC, vitaminc C) or juice (OJ, orange juice). This is a categoric field of values (OJ, VC). There are equal unmber of records (30) for each.
- "dose" shows the dose of intake per day. This is categoric field of values (0.5, 1, 2). Median is 1 indicating uniform distribution

4. Boxplot of "len" IAW "supp" category

```
tg |>
  ggplot(aes(y=len)) +
  geom_boxplot(aes(fill=supp)) +
  labs(title="Teeth Length Distribution by Supplement", x="", y="Teeth Length", fill="supplement")
```



Hypothesis Tests

1. Is there a considerable difference in "len" between "supp" categories: OJ, VC

```
oj_len <- tg[tg$supp=="OJ", "len"]
vc_len <- tg[tg$supp=="VC", "len"]
```

```r
t.test(x=oj_len, y=vc_len)
```

```
##
##  Welch Two Sample t-test
##
## data:  oj_len and vc_len
## t = 1.9153, df = 55.309, p-value = 0.06063
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.1710156  7.5710156
## sample estimates:
## mean of x mean of y
##  20.66333  16.96333
```

```r
# defaults: alt:two sided test, conf.level: 0.95, paired=F, var.equal=F
```

Comments

Assumptions: We seek for difference. Hence two sided test is applied.

1. Calculated positive t value is less than what alpha (0.05) dictates.
2. P value is greater than alpha
3. Confidence interval includes zero

RESULT: Null hypothesis of "means are equal" is FAILED TO REJECT.

2. Is there a considerable difference in "len" between "dose" categories: 0.5, 1, 2

Data manipulation

```r
half_len <- tg[tg$dose==0.5, "len"]
one_len <- tg[tg$dose==1, "len"]
two_len <- tg[tg$dose==2, "len"]
```

Fist t test between the teeth length effect of 0.5 and 1 doses of vitamin c supplement

```r
t.test(x=half_len, y=one_len, alt="less")
```

```
##
##  Welch Two Sample t-test
##
## data:  half_len and one_len
## t = -6.4766, df = 37.986, p-value = 6.342e-08
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##       -Inf -6.753323
## sample estimates:
## mean of x mean of y
##    10.605    19.735
```

```r
# defaults: conf.level: 0.95, paired=F, var.equal=F
```

Second t test between the teeth length effect of 1 and 2 doses of vitamin c supplement

```r
t.test(x=one_len, y=two_len, alt="less")
```

```
##
##  Welch Two Sample t-test
```

```
##
## data:  one_len and two_len
## t = -4.9005, df = 37.101, p-value = 9.532e-06
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##      -Inf -4.17387
## sample estimates:
## mean of x mean of y
##    19.735    26.100
```

```
# defaults: conf.level: 0.95, paired=F, var.equal=F
```

Comments

Assumptions: We assume lesser dose would translate into lesser teeth elongation. Hence one sided (lesser than) test is applied.

1. calculated t values are considerably less than what alpha (0.05) dictates.
2. p values is way lesser than alpha
3. Confidence intervals does not include zero

RESULT: Null hypothesis of "true difference in means is NOT less than 0" is rejected.

3. Is there a considerable difference in "len" between "supp" categories: OJ, VC when the "dose":2

```
tg_two <- tg[tg$dose==2,]

oj_two_len <- tg[tg_two$supp=="OJ", "len"]
vc_two_len <- tg[tg_two$supp=="VC", "len"]

t.test(x=oj_two_len, y=vc_two_len)
```

```
##
##  Welch Two Sample t-test
##
## data:  oj_two_len and vc_two_len
## t = -0.12719, df = 52.685, p-value = 0.8993
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -4.248776  3.742109
## sample estimates:
## mean of x mean of y
##  18.68667  18.94000
```

```
# defaults: alt:two sided test, conf.level: 0.95, paired=F, var.equal=F)
```

Comments

Assumptions: We seek for difference. Hence two sided test is applied.

1. calculated negative t value is considerably more than what alpha (0.05) dictates.
2. p values is way greater than alpha
3. Confidence intervals includes zero

RESULT: Null hypothesis of "true difference in means is 0" is failed to reject.